

the optimization recommendation, Mr. N's proposed asset allocation was 50 percent stocks, 40 percent bonds, and 10 percent cash. Spencer's job is now to answer the following three questions:

1. What effect do Mr. N's biases have on the asset allocation decision?
2. Should Spencer moderate or adapt to Mr. N's biases?
3. What is an appropriate behaviorally modified asset allocation for Mr. N?

Solutions to Mr. Nicholas Case Study

Effect of Biases Mr. Nicholas has emotional biases that provide a clear indication of what allocation he would naturally prefer, which is one dominated by equities. Mr. Nicholas's overconfidence leads him to be more comfortable with equities than may be appropriate. This overconfidence, as well as a potential endowment bias, may explain why he has chosen to hold a substantial part of his portfolio in his employing company's stock. Additionally, Mr. N is likely to experience regret if he misses a major market move. However, he has a high need for current income that supplements his "spend today" mentality (self-control bias), and he may need the "ballast" of fixed-income investments in the event of a market downturn. Because his level of wealth is high, however, he does have some flexibility with his allocation to favor equity over fixed income.

Moderate or Adapt? When considering level of wealth, Mr. Nicholas clearly does not run a standard of living risk, which argues for adapting to his biases. Additionally, his behavioral biases are principally emotional (overconfidence, regret aversion, self-control). Given these facts, and that he naturally prefers an allocation favoring equity, Spencer now has the information with which to make the graph shown in Figure 25.1 and the "moderate, adapt, or both moderate *and* adapt" recommendation.

Spencer decides that the appropriate recommendation is to *adapt* to Mr. N's biases and create a more aggressive portfolio that Mr. N can adhere to and be comfortable with. Spencer also recommends that Mr. N reduce his company stock position by 50 percent and reduce spending, if possible. At the same time, Spencer will run a cash flow analysis to ensure that in the event of a market downturn, Mr. N's living expenses will not be at risk. Spencer also advises a comfortable cash reserve.

Behaviorally Modified Portfolio Decision The mean-variance optimizer's recommended allocation was **50 percent stocks, 40 percent bonds, and 10 percent cash**. Using guidelines presented earlier in the reading for